UNIT –II Data Preprocessing

Data preprocessing:- An overview, Data cleaning, Data Integration, Data Reduction, Data transformation and Data discretization.

Data preprocessing is an important step in the data mining process. It refers to the cleaning, transforming, and integrating of data in order to make it ready for analysis. The goal of data preprocessing is to improve the quality of the data and to make it more suitable for the specific data mining task.

Some common steps in data preprocessing include:

Data Cleaning: This involves identifying and correcting errors or inconsistencies in the data, such as missing values, outliers, and duplicates. Various techniques can be used for data cleaning, such as imputation, removal, and transformation.

Data Integration: This involves combining data from multiple sources to create a unified dataset. Data integration can be challenging as it requires handling data with different formats, structures, and semantics. Techniques such as record linkage and data fusion can be used for data integration.

Data Transformation: This involves converting the data into a suitable format for analysis. Common techniques used in data transformation include normalization, standardization, and discretization. Normalization is used to scale the data to a common range, while standardization is used to transform the data to have zero mean and unit variance. Discretization is used to convert continuous data into discrete categories.

Data Reduction: This involves reducing the size of the dataset while preserving the important information. Data reduction can be achieved through techniques such as feature selection and feature extraction. Feature selection involves selecting a subset of relevant features from the dataset, while feature extraction involves transforming the data into a lower-dimensional space while preserving the important information.

Data Discretization: This involves dividing continuous data into discrete categories or intervals. Discretization is often used in data mining and machine learning algorithms that require categorical data. Discretization can be achieved through techniques such as equal width binning, equal frequency binning, and clustering.

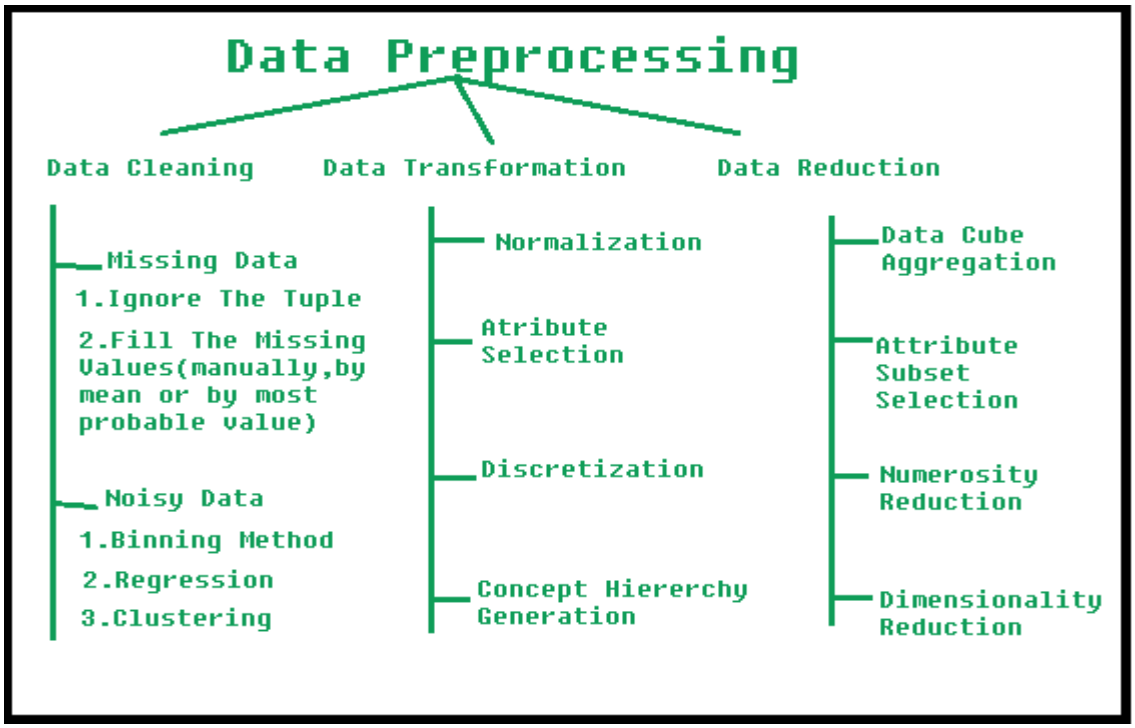
Data Normalization: This involves scaling the data to a common range, such as between 0 and 1 or -1 and 1. Normalization is often used to handle data with different units and scales. Common normalization techniques include min-max normalization, z-score normalization, and decimal scaling.

Data preprocessing plays a crucial role in ensuring the quality of data and the accuracy of the analysis results. The specific steps involved in data preprocessing may vary depending on the nature of the data and the analysis goals.

By performing these steps, the data mining process becomes more efficient and the results become more accurate.

Preprocessing in Data Mining:

Data preprocessing is a data mining technique which is used to transform the raw data in a useful and efficient format.



Steps Involved in Data Preprocessing:

1. Data Cleaning:
The data can have many irrelevant and missing parts. To handle this part, data cleaning is done. It involves handling of missing data, noisy data etc.
(a). Missing Data:
This situation arises when some data is missing in the data. It can be handled in various ways.
Some of them are:

1. **Ignore the tuples:**
   This approach is suitable only when the dataset we have is quite large and multiple values are missing within a tuple.

2. **Fill the Missing values:**
   There are various ways to do this task. You can choose to fill the missing values manually, by attribute mean or the most probable value.

- **(b). Noisy Data:**
  Noisy data is a meaningless data that can't be interpreted by machines.It can be generated due to faulty data collection, data entry errors etc. It can be handled in following ways :
  1. **Binning Method:**
     This method works on sorted data in order to smooth it. The whole data is divided into segments of equal size and then various methods are performed to complete the task. Each segmented is handled separately. One can replace all data in a segment by its mean or boundary values can be used to complete the task.

  2. **Regression:**
     Here data can be made smooth by fitting it to a regression function.The regression used may be linear (having one independent variable) or multiple (having multiple independent variables).

  3. **Clustering:**
     This approach groups the similar data in a cluster. The outliers may be undetected or it will fall outside the clusters.

2. Data Transformation:
This step is taken in order to transform the data in appropriate forms suitable for mining process. This involves following ways:

1. **Normalization:**
   It is done in order to scale the data values in a specified range (-1.0 to 1.0 or 0.0 to 1.0)

2. **Attribute Selection:**
   In this strategy, new attributes are constructed from the given set of attributes to help the mining process.

3. **Discretization:**
   This is done to replace the raw values of numeric attribute by interval levels or conceptual levels.

4. **Concept Hierarchy Generation:**
   Here attributes are converted from lower level to higher level in hierarchy. For Example-The attribute "city" can be converted to "country".

3. Data Reduction:
Data reduction is a crucial step in the data mining process that involves reducing the size of the dataset while preserving the important information. This is done to improve the efficiency of data analysis and to avoid overfitting of the model. Some common steps involved in data reduction are:

Feature Selection: This involves selecting a subset of relevant features from the dataset. Feature selection is often performed to remove irrelevant or redundant features from the dataset. It can be done using various techniques such as correlation analysis, mutual information, and principal component analysis (PCA).

Feature Extraction: This involves transforming the data into a lower-dimensional space while preserving the important information. Feature extraction is often used when the original features are high-dimensional and complex. It can be done using techniques such as PCA, linear discriminant analysis (LDA), and non-negative matrix factorization (NMF).

Sampling: This involves selecting a subset of data points from the dataset. Sampling is often used to reduce the size of the dataset while preserving the important information. It can be done using techniques such as random sampling, stratified sampling, and systematic sampling.

Clustering: This involves grouping similar data points together into clusters. Clustering is often used to reduce the size of the dataset by replacing similar data points with a representative centroid. It can be done using techniques such as k-means, hierarchical clustering, and density-based clustering.

Compression: This involves compressing the dataset while preserving the important information. Compression is often used to reduce the size of the dataset for storage and transmission purposes. It can be done using techniques such as wavelet compression, JPEG compression, and gzip compression.

Why to preprocess data??

The data to be analyzed by data mining techniques are incomplete i.e. it is lacking attribute values or certain attributes of interest or containing only aggregate data.

It may be noisy i.e. containing errors, or outlier values.

It may be inconsistent i.e. containing discrepancies in the department codes used to categorize items.

Incomplete, noisy and inconsistent data are present in large real world databases and datawarehouses.The data may be incomplete for many reasons.

a) Attributes of interest may not always be available such as customer information for sales transaction data.

b) Other data may not be included simply, because it was not considered important at the time of entry.

c) Relevant data may not be recorded due to a misunderstanding or because of equipment malfunctions.

d) d) Data that were inconsistent with other recorded data may have been deleted.

e) e)The recording of the history or modifications to the data may have been overlooked.

f) Missing data for tuples with missing values for some attributes may need to be inferred.

There are many reasons for noisy data.

a) The instruments used for data collection may be faulty.

b) There may be a human or computer error occurring at data entry.

c) Errors in data transmission can also occur.

d) There may be technology limitations ,such as limited buffer size for coordinating synchronized data transfer and consumption.

e) The incorrect data may be from inconsistencies in naming conventions or data codes used or inconsistent formats for input fields such as “date". Duplicate tuples also require data cleaning.

Data Cleaning

Data cleaning is an essential step in the data mining process. It is crucial to the construction of a model. The step that is required, but frequently overlooked by everyone, is data cleaning. The major problem with quality information management is data quality. Problems with data quality can happen at any place in an information system. Data cleansing offers a solution to these issues.

Data cleaning is the process of correcting or deleting inaccurate, damaged, improperly formatted, duplicated, or insufficient data from a dataset. Even if results and algorithms appear to be correct, they are unreliable if the data is inaccurate. There are numerous ways for data to be duplicated or incorrectly labeled when merging multiple data sources.

In general, data cleaning lowers errors and raises the caliber of the data. Although it might be a time-consuming and laborious operation, fixing data mistakes and removing incorrect information must be done. A crucial method for cleaning up data is data mining. A method for finding useful information in data is data mining. Data quality mining is a novel methodology that uses data mining methods to find and fix data quality issues in sizable databases. Data mining mechanically pulls intrinsic and hidden information from large data sets. Data cleansing can be accomplished using a variety of data mining approaches.

To arrive at a precise final analysis, it is crucial to comprehend and improve the quality of your data. To identify key patterns, the data must be prepared. Exploratory data mining is understood. Before doing business analysis and gaining insights, data cleaning in data mining enables the user to identify erroneous or missing data.

Data cleaning before data mining is often a time-consuming procedure that necessitates IT personnel to assist in the initial step of reviewing your data due to how time-consuming data cleaning is. But if your final analysis is inaccurate or you get an erroneous result, it's possible due to poor data quality.

Steps for Cleaning Data

You can follow these fundamental stages to clean your data even if the techniques employed may vary depending on the sorts of data your firm stores:

1. Remove duplicate or irrelevant observations

Remove duplicate or pointless observations as well as undesirable observations from your dataset. The majority of duplicate observations will occur during data gathering. Duplicate data can be produced when you merge data sets from several sources, scrape data, or get data from clients or other departments. One of the most important factors to take into account in this procedure is de-duplication. Those observations are deemed irrelevant when you observe observations that do not pertain to the particular issue you are attempting to analyze.

You might eliminate those useless observations, for instance, if you wish to analyze data on millennial clients but your dataset also includes observations from earlier generations. This can improve the analysis's efficiency, reduce deviance from your main objective, and produce a dataset that is easier to maintain and use.

2. Fix structural errors

When you measure or transfer data and find odd naming practices, typos, or wrong capitalization, such are structural faults. Mislabelled categories or classes may result from these inconsistencies. For instance, "N/A" and "Not Applicable" might be present on any given sheet, but they ought to be analyzed under the same heading.

3. Filter unwanted outliers

There will frequently be isolated findings that, at first glance, do not seem to fit the data you are analyzing. Removing an outlier if you have a good reason to, such as incorrect data entry, will improve the performance of the data you are working with.

However, occasionally the emergence of an outlier will support a theory you are investigating. And just because there is an outlier, that doesn't necessarily indicate it is inaccurate. To determine the reliability of the number, this step is necessary. If an outlier turns out to be incorrect or unimportant for the analysis, you might want to remove it.

4. Handle missing data

Because many algorithms won't tolerate missing values, you can't overlook missing data. There are a few options for handling missing data. While neither is ideal, both can be taken into account, for example:

Although you can remove observations with missing values, doing so will result in the loss of information, so proceed with caution.

Again, there is a chance to undermine the integrity of the data since you can be working from assumptions rather than actual observations when you input missing numbers based on other observations.

To browse null values efficiently, you may need to change the way the data is used.

5. Validate and QA

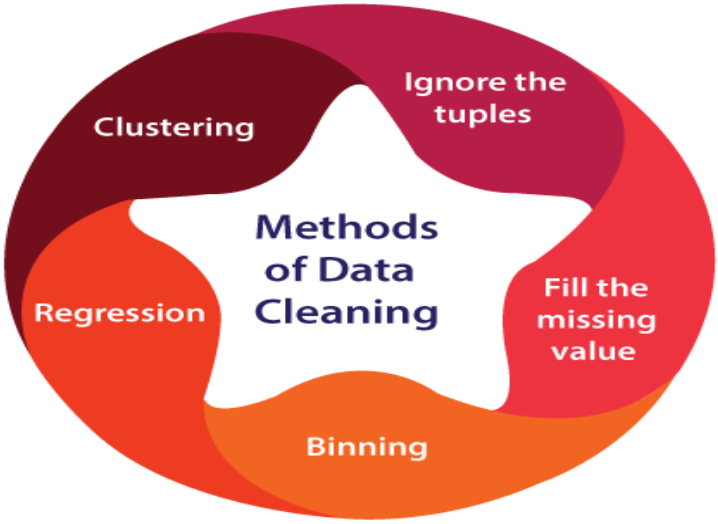
As part of fundamental validation, you ought to be able to respond to the following queries once the data cleansing procedure is complete:

- Are the data coherent?
- Does the data abide by the regulations that apply to its particular field?
- Does it support or refute your working theory? Does it offer any new information?
- To support your next theory, can you identify any trends in the data?
- If not, is there a problem with the data's quality?

False conclusions can be used to inform poor company strategy and decision-making as a result of inaccurate or noisy data. False conclusions can result in a humiliating situation in a reporting meeting when you find out your data couldn't withstand further investigation. Establishing a culture of quality data in your organization is crucial before you arrive. The tools you might employ to develop this plan should be documented to achieve this.

Techniques for Cleaning Data

The data should be passed through one of the various data-cleaning procedures available. The procedures are explained below:



1. **Ignore the tuples:** This approach is not very practical because it is only useful when a tuple has multiple characteristics and missing values.
2. **Fill in the missing value:** This strategy is also not very practical or effective. Additionally, it could be a time-consuming technique. One must add the missing value to the approach. The most common method for doing this is manually, but other options include using attribute means or the most likely value.
3. **Binning method:** This strategy is fairly easy to comprehend. The values nearby are used to smooth the sorted data. The information is subsequently split into several equal-sized parts. The various techniques are then used to finish the assignment.
4. **Regression:** With the use of the regression function, the data is smoothed out. Regression may be multivariate or linear. Multiple regressions have more independent variables than linear regressions, which only have one.
5. **Clustering:** This technique focuses mostly on the group. Data are grouped using clustering. After that, clustering is used to find the outliers. After that, the comparable values are grouped into a "group" or "cluster".

Process of Data Cleaning

The data cleaning method for data mining is demonstrated in the subsequent sections.

1. **Monitoring the errors:** Keep track of the areas where errors seem to occur most frequently. It will be simpler to identify and maintain inaccurate or corrupt information. Information is particularly important when integrating a potential substitute with current management software.

2. **Standardize the mining process:** To help lower the likelihood of duplicity, standardize the place of insertion.
3. **Validate data accuracy:** Analyse the data and spend money on data cleaning software. Artificial intelligence-based tools were utilized to thoroughly check for accuracy.
4. **Scrub for duplicate data:** To save time when analyzing data, find duplicates. By analyzing and investing in independent data-erasing technologies that can analyze imperfect data in quantity and automate the operation, it is possible to avoid again attempting the same data.
5. **Research on data:** Our data needs to be vetted, standardized, and duplicate-checked before this action. There are numerous third-party sources, and these vetted and approved sources can extract data straight from our databases. They assist us in gathering the data and cleaning it up so that it is reliable, accurate, and comprehensive for use in business decisions.
6. **Communicate with the team:** Keeping the group informed will help with client development and strengthening as well as giving more focused information to potential clients.

Data Cleansing Tools can be very helpful if you are not confident of cleaning the data yourself or have no time to clean up all your data sets. You might need to invest in those tools, but it is worth the expenditure. There are many data cleaning tools in the market. Here are some top-ranked data cleaning tools, such as:

1. OpenRefine
2. Trifacta Wrangler
3. Drake
4. Data Ladder
5. Data Cleaner
6. Cloudingo
7. Reifier
8. IBM Infosphere Quality Stage
9. TIBCO Clarity
10. Winpure

<u>Noisy data</u>:- Noise is a random error or variance in a measured variable .Given a numerical attribute such as price ,how can we “smooth” out the data to remove the noise?

Example :- sorted data for price(in dollars):-4,8,15,21,21,24,25,28,34.

Binning methods for data smoothing

Partition into (equal frequency) bins:-

Bin1 : 4,8,15.

Bin2 : 21,21,24.

Bin3 : 25,28,34.

Smoothing by bin means:-

Bin1 : 9,9,9. --(4+8+15=27/3 gives 9)

Bin2 : 22,22,22. --(21+21+24=66/3 gives 22)

Bin3 : 29,29,29. –(25+28+34=87/3 gives 29).

Smoothing by bin boundaries:-

Bin1: 4,4,15.

Bin2: 21,21,24.

Bin3: 25,25,34.

Binning :- This method smooth a sorted data value by consulting its “neighborhood” i.e. values around it. As binning methods consult the neighborhood of values they perform local smoothing.

In above example ,the data for price are first sorted and then partitioned into equal –frequency bins of size 3(i.e. each bin contains 3 values).In smoothing by bin means each value in a bin is replaced by the mean value of the bin.

For example:- from the list 4,8,15,21,21,24,25, 28,34.

The mean of values 4,8,15 in bin1 is(4+8+15=27/3=9).Therefore each original values in this bin is replaced by the value 9.Similarly ,smoothing by bin medians can be employed in which each bin value is replaced by bin median.

Regression:- Data can be smoothed by fitting the data to a function such as with regression. Regression is **a statistical technique that relates a dependent variable to one or more independent variables**. A regression model is able to show whether changes observed in the dependent variable are associated with changes in one or more of the independent variables.

In linear regression we find the “best” line to fit two attributes(or variables)so that one attribute can be used to predict the other. Formulating a regression analysis helps you predict the effects of the independent variable on the dependent one.

Example: we can say that **age and height can be described using a linear regression model**. Since a person's height increases as age increases, they have a linear relationship.

Multiple linear regression:- It is an extension of linear regression ,where more than two attributes are involved and the data are fit to a multidimensional surface.

3) Clustering:- Values that fall outside of the set of clusters are called outliers. clustering is used to detect outliers.

Discrepancy detection:- This is the first step in data cleaning as a process. Discrepancies can be caused by several factors like, poorly designed data entry forms having many optional fields.Human error in data entry.Deliberate errors(e.g. respondents not wanting to give information about themselves) and Data decay (e.g. outdated addresses).

Discrepancies are also due to inconsistent data representations and the inconsistent use of codes.Error in instrumentation devices that record data and system errors.When the data are inadequately used for purpose it is not intended for then also errors occur.Inconsistencies can be due to data integration (e.g. when a given attribute can have different names in different databases).

To detect discrepancies use any knowledge regarding properties of the data, such knowledge or “data about data” is called metadata.

Look for the inconsistent use of codes and any inconsistent data representation (such as “2024/07/25” and “25/07/2024” for date).

- Unique rule:- This rule says that each value of the given attribute must be different from all other values for that attribute.(i.e. all tuples for that attribute should have different values no repeated values allowed.).
- Consecutive rule:-This rule says that there can be no missing values between the lowest and the highest values for the attribute, and that all values must also be unique.
- Null rule:- This rule specifies the use of blanks ,question marks, special characters or other strings that may indicate the null condition. The null rule should specify how to record the null condition .The data should be examined regarding unique rules , consecutive rules and null rules
- Example:- 1) To store zero(0) for numerical attributes.
- 2) A blank for character attributes or any other convention that may in use (such as “don’t know” or “?”)should be transformed to blank
- For example:- A salesman in a firm or company will have certain commission value, rest all other employees will have commission as NULL. It should be remembered that NULL is different from zero(0).
- Some of the reasons for missing values are:-
- 1) The person originally asked to provide a value for the attribute refuses to fill that value and/or finds that the information requested is not applicable.(e.g. driving license number left blank by individuals who are not drivers).
- 2) The data entry person does not know the correct value.

3)The value is to be provided by a later step of the process

- Different commercial tools available for discrepancy detection are as follows:-
- Data Scrubbing Tools:- They use simple domain knowledge(e.g. knowledge of postal addresses and spell checking) to detect errors and make corrections in the data. These rules rely on parsing and fuzzy matching techniques when cleaning data from multiple sources.
- Data Auditing Tools:- They find discrepancies by analyzing the data to discover rules and relationships and detecting data that violate such conditions. Some data inconsistencies may be corrected manually using external references.
- For example: Errors made at data entry may be corrected by performing a paper trace. Most errors will require data transformation which is a second step in data cleaning process. i.e. once the discrepancies are known we need to define and apply (a series of)transformations to correct them.

Commercial tools can assist in the data transformation step .Data migration tools allow simple transformations to be specified ,such as to replace the string “gender” by “sex”.

- The two step process of discrepancy detection and data transformation (to correct discrepancies) iterates. This process is error prone and time consuming.
- Some transformations may introduce more discrepancies. some nested discrepancies may only be detected after others have been fixed.
- Any tuples that cannot be automatically handled by a given transformation are written to a file without any explanation regarding the reason behind their failure. The entire data cleaning process also suffers from the lack of interactivity
- A publicly available data cleaning tool is **“*potter’s wheel*”.**
- It integrates discrepancy detection and transformation.
- The tool performs discrepancy checking automatically in the background on the latest transformed view of the data.

Users can gradually develop and refine transformations, as discrepancies are found ,this leads to more effective and efficient data cleaning

Data Integration:-

Data integration in data mining refers to the process of combining data from multiple sources into a single, unified view. This can involve cleaning and transforming the data, as well as resolving any inconsistencies or conflicts that may exist between the different sources. The goal of data integration is to make the data more useful and meaningful for the purposes of analysis and decision making. Techniques used in data integration include data warehousing, ETL (extract, transform, load) processes, and data federation.

Data Integration is a data preprocessing technique that combines data from multiple heterogeneous data sources into a coherent data store and provides a unified view of the data. These sources may include multiple data cubes, databases, or flat files.

Data integration is the process of combining data from multiple sources into a cohesive and consistent view. This process involves identifying and accessing the different data sources, mapping the data to a common format, and reconciling any inconsistencies or discrepancies between the sources. The goal of data integration is to make it easier to access and analyze data that is spread across multiple systems or platforms, in order to gain a more complete and accurate understanding of the data.

Data integration can be challenging due to the variety of data formats, structures, and semantics used by different data sources. Different data sources may use different data types, naming conventions, and schemas, making it difficult to combine the data into a single view. Data integration typically involves a combination of manual and automated processes, including data profiling, data mapping, data transformation, and data reconciliation.

Issues in Data Integration:

There are several issues that can arise when integrating data from multiple sources, including:

1. **Data Quality:** Inconsistencies and errors in the data can make it difficult to combine and analyze.
2. **Data Semantics:** Different sources may use different terms or definitions for the same data, making it difficult to combine and understand the data.
3. **Data Heterogeneity:** Different sources may use different data formats, structures, or schemas, making it difficult to combine and analyze the data.
4. **Data Privacy and Security:** Protecting sensitive information and maintaining security can be difficult when integrating data from multiple sources.
5. **Scalability:** Integrating large amounts of data from multiple sources can be computationally expensive and time-consuming.
6. **Data Governance:** Managing and maintaining the integration of data from multiple sources can be difficult, especially when it comes to ensuring data accuracy, consistency, and timeliness.
7. **Performance:** Integrating data from multiple sources can also affect the performance of the system.
8. **Integration with existing systems:** Integrating new data sources with existing systems can be a complex task, requiring significant effort and resources.
9. **Complexity:** The complexity of integrating data from multiple sources can be high, requiring specialized skills and knowledge.

There are three issues to consider during data integration:

1) **Schema Integration.**
2) **Redundancy Detection.**
3) **Resolution of data value conflicts.**

These are explained in brief below.

1. Schema Integration:

- Integrate metadata from different sources.
- The real-world entities from multiple sources are referred to as the entity identification problem.ER
- Schema Integration:- Schema integration and object matching can be tricky. How to match equivalent real-world entities from multiple data sources?. This is called entity identification problem.
- For example-how can the data analyst or computer be sure that customer_id in one database and cust_number in another, refer to the same attribute?
- Examples of metadata for each attribute include the name, meaning , data type and range of values permitted for the attribute, and null rules for handling blank , zero or Null values.

2. Redundancy Detection:

- An attribute may be redundant if it can be derived or obtained from another attribute or set of attributes.
- Inconsistencies in attributes can also cause redundancies in the resulting data set.
- Some redundancies can be detected by correlation analysis.
- Redundancy:-An attribute(such as annual revenue) may be redundant if it can be "derived" from another attribute or set of attributes.
- Inconsistencies in attribute or dimension naming can also cause redundancies in the resulting data set.
- Some redundancies can be detected by correlation analysis. when 2 attributes are given , correlation analysis measures how strongly one attribute implies the other based on the available data.
- For numerical attributes ,we can evaluate the correlation between two attributes A and B, by computing the correlation coefficient. the higher the value ,the stronger is the correlation.
- In addition to detecting redundancies between attributes ,duplication should also be detected at the tuple level.
- The use of de normalized tables also causes redundancy.
- So to avoid redundancy always use normalized tables.

3. Resolution of data value **conflicts:**

- This is the third critical issue in data integration.
- Attribute values from different sources may differ for the same real-world entity.
- An attribute in one system may be recorded at a lower level of abstraction than the "same" attribute in another.
- For the same real world entity ,attribute values from different sources may be different .This may due to differences in representation ,scaling or encoding.
- For example:- a weight attribute may be stored in metric units in one system and british imperial units in another system.
- An attribute in one system may be recorded at a lower-level of abstraction than the same attribute in another system.
- When matching attributes from one database to another during integration ,special attention must be paid to the structure of data. This is done to ensure that any attribute function dependencies and referential constraints in the source system match those in the target system.
- Careful integration of the data from multiple sources can help reduce and avoid redundancies and inconsistencies in the resulting data set.
- This can help to improve the accuracy and speed of the subsequent mining process.

Data Transformation :-

Data transformation in data mining refers to the process of converting raw data into a format that is suitable for analysis and modeling. The goal of data transformation is to prepare the data for data mining so that it can be used to extract useful insights and knowledge. Data transformation typically involves several steps, including:

1. **Data cleaning:** Removing or correcting errors, inconsistencies, and missing values in the data.
2. **Data integration:** Combining data from multiple sources, such as databases and spreadsheets, into a single format.
3. **Data normalization:** Scaling the data to a common range of values, such as between 0 and 1, to facilitate comparison and analysis.
4. **Data reduction:** Reducing the dimensionality of the data by selecting a subset of relevant features or attributes.
5. **Data discretization**: Converting continuous data into discrete categories or bins.
6. **Data aggregation:** Combining data at different levels of granularity, such as by summing or averaging, to create new features or attributes.
7. Data transformation is an important step in the data mining process as it helps to ensure that the data is in a format that is suitable for analysis and modeling, and that it is free of errors and inconsistencies. Data transformation can also help to improve the performance of data mining algorithms, by reducing the dimensionality of the data, and by scaling the data to a common range of values.

The data are transformed in ways that are ideal for mining the data. The data transformation involves steps that are:

1. Smoothing: It is a process that is used to remove noise from the dataset using some algorithms It allows for highlighting important features present in the dataset. It helps in predicting the patterns. When collecting data, it can be manipulated to eliminate or reduce any variance or any other noise form. The concept behind data smoothing is that it will be able to identify simple changes to help predict different trends and patterns. This serves as a help to analysts or traders who need to look at a lot of data which can often be difficult to digest for finding patterns that they wouldn't see otherwise.

2. Aggregation: Data collection or aggregation is the method of storing and presenting data in a summary format. The data may be obtained from multiple data sources to integrate these data sources into a data analysis description. This is a crucial step since the accuracy of data analysis insights is highly dependent on the quantity and quality of the data used. Gathering accurate data of high quality and a large enough quantity is necessary to produce relevant results. The collection of data is useful for everything from decisions concerning financing or business strategy of the product, pricing, operations, and marketing strategies. For **example**, Sales, data may be aggregated to compute monthly& annual total amounts.

3. Discretization: It is a process of transforming continuous data into set of small intervals. Most Data Mining activities in the real world require continuous attributes. Yet many of the existing data mining frameworks are unable to handle these attributes. Also, even if a data mining task can manage a continuous attribute, it can significantly improve its efficiency by replacing a constant quality attribute with its discrete values. For **example**, (1-10, 11-20) (age:- young, middle age, senior).

4. Attribute Construction: Where new attributes are created & applied to assist the mining process from the given set of attributes. This simplifies the original data & makes the mining more efficient.

5. Generalization: It converts low-level data attributes to high-level data attributes using concept hierarchy. For Example Age initially in Numerical form (22, 25) is converted into categorical value (young, old). For **example**, Categorical attributes, such as house addresses, may be generalized to higher-level definitions, such as town or country.

6. Normalization: Data normalization involves converting all data variables into a given range. Techniques that are used for normalization are:

- **Min-Max Normalization:**
  - This transforms the original data linearly.
  - Suppose that: min_A is the minima and max_A is the maxima of an attribute, P
  - Where v is the value you want to plot in the new range.
  - v' is the new value you get after normalizing the old value.
  - The min-max normalization maps a value ,V,of A to V' in the range[new_minA , new_maxA] by computing.
  - V'=V-minA/maxA-minA(new_maxA– new_minA)+new_minA.
  - $$\boxed{v' = \frac{v - \min(A)}{\max(A) - \min(A)}(\text{new_max}(A) - \text{new_min}(A)) + \text{new_min}(A)}$$
  -
- **Z-Score Normalization:**
  - In z-score normalization (or zero-mean normalization) the values of an attribute (A), are normalized based on the mean of A and its standard deviation
  - A value, v, of attribute A is normalized to v' by computing.
  - V'=V-(mean of A)/standard deviation of A.
  - $$\boxed{v' = \frac{v - \overline{A}}{\sigma_A}}$$
- **Decimal Scaling:**
  - It normalizes the values of an attribute by changing the position of their decimal points
  - The number of points by which the decimal point is moved can be determined by the absolute maximum value of attribute A.
  - A value, v, of attribute A is normalized to v' by computing
  - where j is the smallest integer such that Max(|v'|) < 1.
  - Suppose: Values of an attribute P varies from -99 to 99.
  - The maximum absolute value of P is 99.
  - For normalizing the values we divide the numbers by 100 (i.e., j = 2) or (number of integers in the largest number) so that values come out to be as 0.98, 0.97 and so on.
  - It normalizes by moving the decimal point of values of attribute A. The number of decimal points moved depends on the maximum absolute value of A. The value ,V, of A is normalized to V' by computing,
  - $V'=V/10^j$ where j is the smallest integer such that Max(|V'|<1)

Data Reduction

Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information. This can be beneficial in situations where the dataset is too large to be processed efficiently, or where the dataset contains a large amount of irrelevant or redundant information.

There are several different data reduction techniques that can be used in data mining, including:

1. **Data Sampling:** This technique involves selecting a subset of the data to work with, rather than using the entire dataset. This can be useful for reducing the size of a dataset while still preserving the overall trends and patterns in the data.
2. **Dimensionality Reduction:** This technique involves reducing the number of features in the dataset, either by removing features that are not relevant or by combining multiple features into a single feature.
3. **Data Compression:** This technique involves using techniques such as lossy or lossless compression to reduce the size of a dataset.
4. **Data Discretization:** This technique involves converting continuous data into discrete data by partitioning the range of possible values into intervals or bins.

5. **Feature Selection:** This technique involves selecting a subset of features from the dataset that are most relevant to the task at hand.
6. It's important to note that data reduction can have a trade-off between the accuracy and the size of the data. The more data is reduced, the less accurate the model will be and the less generalizable it will be.

Methods of data reduction:
These are explained as following below.

1. Data Cube Aggregation:
This technique is used to aggregate data in a simpler form. For example, imagine the information you gathered for your analysis for the years 2012 to 2014, that data includes the revenue of your company every three months. They involve you in the annual sales, rather than the quarterly average, So we can summarize the data in such a way that the resulting data summarizes the total sales per year instead of per quarter. It summarizes the data.

2. Dimension reduction:
Whenever we come across any data which is weakly important, then we use the attribute required for our analysis. It reduces data size as it eliminates outdated or redundant features.

- **Step-wise Forward Selection –**
  The selection begins with an empty set of attributes later on we decide the best of the original attributes on the set based on their relevance to other attributes. We know it as a p-value in statistics.

  Suppose there are the following attributes in the data set in which few attributes are redundant.

```
Initial attribute Set: {X1, X2, X3, X4, X5, X6}
Initial reduced attribute set:  { }

Step-1: {X1}
Step-2: {X1, X2}
Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}
```

- **Step-wise Backward Selection –**
  This selection starts with a set of complete attributes in the original data and at each point, it eliminates the worst remaining attribute in the set.

  Suppose there are the following attributes in the data set in which few attributes are redundant.

```
Initial attribute Set: {X1, X2, X3, X4, X5, X6}
Initial reduced attribute set:  {X1, X2, X3, X4, X5, X6 }

Step-1: {X1, X2, X3, X4, X5}
Step-2: {X1, X2, X3, X5}
Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}
```

- **Combination of forwarding and Backward Selection –**
  It allows us to remove the worst and select the best attributes, saving time and making the process faster.

3. <u>Data Compression</u>:
The data compression technique reduces the size of the files using different encoding mechanisms (Huffman Encoding & run-length Encoding). We can divide it into two types based on their compression techniques.

- **Lossless Compression –**
  Encoding techniques (Run Length Encoding) allow a simple and minimal data size reduction. Lossless data compression uses algorithms to restore the precise original data from the compressed data.
- **Lossy Compression –**
  Methods such as the Discrete Wavelet transform technique, PCA (principal component analysis) are examples of this compression. For e.g., the JPEG image format is a lossy compression, but we can find the meaning equivalent to the original image. In lossy-data compression, the decompressed data may differ from the original data but are useful enough to retrieve information from them.

5. **Numerosity Reduction:**
   In this reduction technique, the actual data is replaced with mathematical models or smaller representations of the data instead of actual data, it is important to only store the model parameter. Or non-parametric methods such as clustering, histogram, and sampling.
   The data volume can be reduced by choosing alternative smaller forms of data representation.For this purpose numerosity reduction can be applied.These techniques can be parametric and non parametric.
   Parametric method:- A model is used to estimate the data ,so that only the data parameters are stored, instead of actual data.

Example:- Log linear models

4b)**Non-Parametric method**:-These methods for storing reduced representation of data are non parametric methods.

Example:- Histograms, clustering and sampling.

** **Log linear models**:-They approximate discrete multidimensional probability distribution .Log linear models are used to estimate the probability of each point in a multidimensional space for a set of discretized attributes based on a smaller subset of dimensional combinations. This allows a higher dimensional data space to be constructed from lower dimensional spaces. Log linear models are also useful for dimensionality reduction and data smoothing.

4b)**Non-Parametric method**:-, clustering and sampling.

Histograms:-They are a popular method of data reduction. It uses binning to approximate data distribution. The histogram for an attribute **A, partitions the data distribution of **A** into disjoint subsets or buckets. If each bucket represents only a single attribute-value/frequency pair the buckets are called singleton buckets.

Histograms:-There are several partitioning rules to determine buckets and partition attribute values.

i)Equal width:- In equal width histogram ,the width of each bucket range is uniform.

ii) Equal frequency (or equidepth):- The buckets are created so that, the frequency of each bucket is constant(i.e. each bucket contains the same number of contiguous data samples).

iii)V-optimal:- This histogram has the least variance. Histogram variance is a weighted sum of the original values that each bucket represents, where bucket weight is equal to the number of values in the bucket.

iv)MaxDiff:- In this histogram we consider the difference between each pair of adjacent values. A bucket boundary is established between each pair ,for pairs having the B-1 largest differences ,where B is the number of buckets specified by user.

5. Discretization & Concept Hierarchy Operation:
Techniques of data discretization are used to divide the attributes of the continuous nature into data with intervals. We replace many constant values of the attributes by labels of small intervals. This means that mining results are shown in a concise, and easily understandable way.

- **Top-down discretization –**
  If you first consider one or a couple of points (so-called breakpoints or split points) to divide the whole set of attributes and repeat this method up to the end, then the process is known as top-down discretization also known as splitting.
- **Bottom-up discretization –**
  If you first consider all the constant values as split points, some are discarded through a combination of the neighborhood values in the interval, that process is called bottom-up discretization.

Concept Hierarchies:
It reduces the data size by collecting and then replacing the low-level concepts (such as 43 for age) with high-level concepts (categorical variables such as middle age or Senior).

For numeric data following techniques can be followed:

- **Binning –**
  Binning is the process of changing numerical variables into categorical counterparts. The number of categorical counterparts depends on the number of bins specified by the user.
- **Histogram analysis –**
  Like the process of binning, the histogram is used to partition the value for the attribute X, into disjoint ranges called brackets. There are several partitioning rules:
  1. **Equal Frequency partitioning:** Partitioning the values based on their number of occurrences in the data set.
  2. **Equal Width Partitioning:** Partitioning the values in a fixed gap based on the number of bins i.e. a set of values ranging from 0-20.
  3. **Clustering:** Grouping similar data together.

Discretization in data mining

Data discretization refers to a method of converting a huge number of data values into smaller ones so that the evaluation and management of data become easy. In other words, data discretization is a method of converting attributes values of continuous data into a finite set of intervals with minimum data loss. There are two forms of data discretization first is supervised discretization, and the second is unsupervised discretization. Supervised discretization refers to a method in which the class data is used. Unsupervised discretization refers to a method depending upon the way which operation proceeds. It means it works on the top-down splitting strategy and bottom-up merging strategy.

Now, we can understand this concept with the help of an example

Suppose we have an attribute of Age with the given values

Age 1,5,9,4,7,11,14,17,13,18, 19,31,33,36,42,44,46,70,74,78,77

Table before Discretization

Attribute	Age	Age	Age	Age
	1,5,4,9,7	11,14,17,13,18,19	31,33,36,42,44,46	70,74,77,78
After Discretization	Child	Young	Mature	Old

Some Famous techniques of data discretization

Histogram analysis

Histogram refers to a plot used to represent the underlying frequency distribution of a continuous data set. Histogram assists the data inspection for data distribution. For example, Outliers, skewness representation, normal distribution representation, etc.

Binning

Binning refers to a data smoothing technique that helps to group a huge number of continuous values into smaller values. For data discretization and the development of idea hierarchy, this technique can also be used.

Cluster Analysis

Cluster analysis is a form of data discretization. A clustering algorithm is executed by dividing the values of x numbers into clusters to isolate a computational feature of x.

Data discretization using decision tree analysis

Data discretization refers to a decision tree analysis in which a top-down slicing technique is used. It is done through a supervised procedure. In a numeric attribute discretization, first, you need to select the attribute that has the least entropy, and then you need to run it with the help of a recursive process. The recursive process divides it into various discretized disjoint intervals, from top to bottom, using the same splitting criterion.

Data discretization using correlation analysis

Discretizing data by linear regression technique, you can get the best neighboring interval, and then the large intervals are combined to develop a larger overlap to form the final 20 overlapping intervals. It is a supervised procedure.

Data discretization and concept hierarchy generation

The term hierarchy represents an organizational structure or mapping in which items are ranked according to their levels of importance. In other words, we can say that a hierarchy concept refers to a sequence of mappings with a set of more general concepts to complex concepts. It means mapping is done from low-level concepts to high-level concepts. For example, in computer science, there are different types of hierarchical systems. A document is placed in a folder in windows at a specific place in the tree structure is the best example of a computer hierarchical tree model. There are two types of hierarchy: top-down mapping and the second one is bottom-up mapping.

Let's understand this concept hierarchy for the dimension location with the help of an example.

A particular city can map with the belonging country. For example, New Delhi can be mapped to India, and India can be mapped to Asia.

Top-down mapping

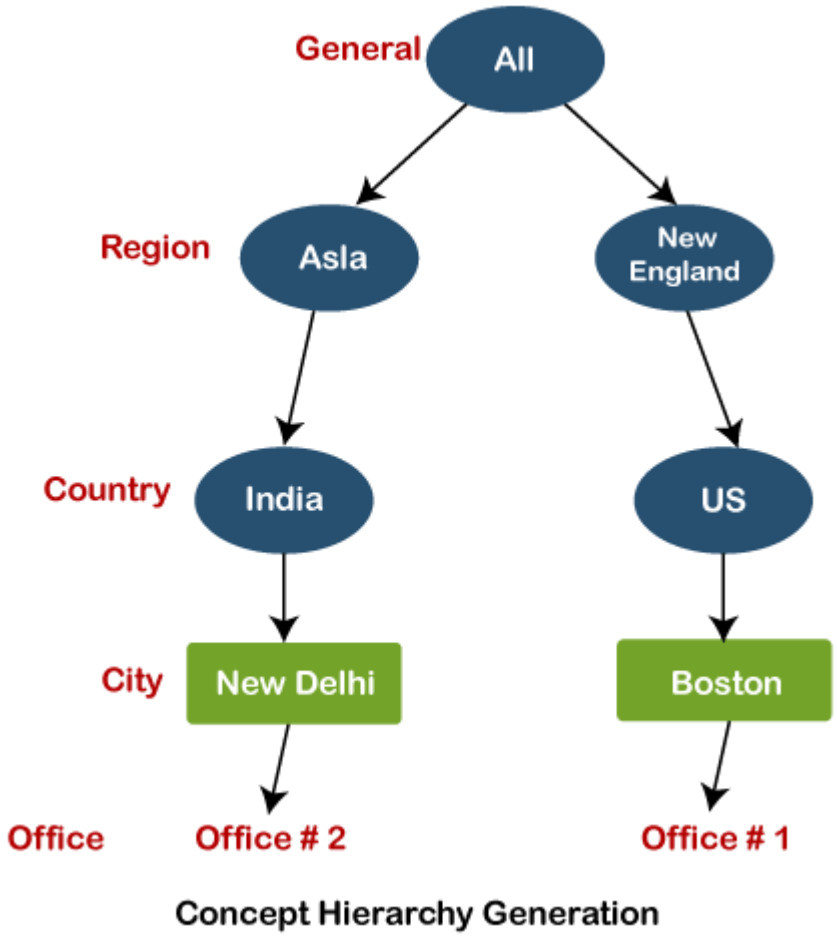
Top-down mapping generally starts with the top with some general information and ends with the bottom to the specialized information.

Bottom-up mapping

Bottom-up mapping generally starts with the bottom with some specialized information and ends with the top to the generalized information.

Bottom-up mapping

Bottom-up mapping generally starts with the bottom with some specialized information and ends with the top to the generalized information.



**********END OF UNIT 2**********